

Improving the Reliability of LLM-based Financial Advisors Through Hybrid Verification Systems

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March 2026

1. Introduction

During the midterm literature review, I studied the evolution of LLM financial advice systems from rule-based systems, to robo-advisors, to the emergence of LLM-based financial systems. The midterm paper concluded that LLM-based systems do represent a paradigm shift for digital financial advisors, and that due to their ability to retain large amounts of context and personal information, combined with RAG and ability to search the web, these types of systems could revolutionize how financial literacy is taught to young adults and those looking to better manage their finances. However, one of the main limitations that the paper also found was that LLM-based systems can hallucinate and provide incorrect or false information. That, combined with the high-stakes nature of finance, presents a great risk but also an opportunity to build something better.

2. What I Want to Study

I want to study how we can make LLM-based systems safer and less likely to hallucinate in financial advice contexts. Specifically, I want to study the impact of adding a “verification” layer between the raw LLM output and the output shown to the user. This verification layer would draw inspiration from rule-based financial systems and incorporate structured sources and documentation to validate generated responses. The proposed system would take the best of both worlds: the conversational ability and personalization of LLM-based systems and the reliability and grounded-in-truth approach of rules-based systems. When a user asks a financial question, the LLM would first generate a response using its standard reasoning. Before that response is shown to the user, however, the verification layer would analyze the output and compare it against a set of trusted financial sources such as financial literacy

datasets, publicly available information, or predefined financial rules. If the system detects a hallucination, it would flag the answer with a warning and then either correct the output using retrieved sources or request that the LLM generate a new answer without any prior context.

Project Goal

The primary goal of this project is to evaluate the effectiveness of hybrid systems that combine LLM-based generative AI with rules-based verification mechanisms against pure LLM-only systems and determine whether the former improves reliability. To do so, we will use publicly available and synthetically generated datasets to simulate the types of questions that a digital financial advisor would be asked. We will evaluate the models' performance based on answer accuracy, hallucination rate, and the verification layer's ability to detect and correct incorrect financial advice.

Why This Is Forward-Looking

The forward-looking aspect of this project is the hybrid combination of a rules-based system design and an LLM agent. To the best of my knowledge, this is one of the first attempts to design and evaluate a system that combines structured financial verification with genAI outputs and directly compares it against pure LLM-only systems. The results of this experiment will contribute to understanding how financial AI systems might be safely deployed in real-world settings. Additionally, the results on the hybrid system's efficacy compared to the control (the LLM-only model) will show whether a verification-based system is the answer to the reliability problem posed by LLMs and hallucinations in high-risk environments, beyond just financial systems.

Research Questions

- RQ1: How frequently do LLM-based financial advice systems hallucinate or produce incorrect information when answering common/advanced financial questions?
- RQ2: Can a verification layer that references structured financial rules and trusted documents improve response accuracy compared to a control LLM-only model?
- RQ3: What are the best practices for implementing a hybrid model like this that combines LLM outputs with rules-based verification?

3. Evaluation and Results

To evaluate the proposed system, I will create a dataset of multiple-choice personal finance questions. This dataset will use publicly available financial literacy tests and open-source financial quizzes available on Github, as well as synthetically generated queries that simulate advanced questions for a digital financial advisor. I will then create an evaluation framework that asks for both the types of systems for each question and records their outputs. The responses will be evaluated for correctness by comparing them against the correct answer (in the case of a public dataset) or by assessing their closeness to the answer (for a synthetic dataset, in case there are open-ended questions too). Performance for each model will be evaluated using answer accuracy, hallucination rate, and error detection rate (for the validation layer). This will be used to measure how often incorrect advice is given and how effectively the verification layer catches and fixes hallucinations.